AE462 Lab 5: Unsteady Flow Fields

Lab Session: Monday, March 1, 2026
Due: Monday, March 25, 2026, 9am CST

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Introduction

This laboratory investigates unsteady flow behavior of air over a cylinder using CFD simulations at various Reynolds numbers. The focus is on identifying transient behavior, analyzing periodic vortex shedding, and extracting dominant frequencies.

I. Transient Time Analysis

A. Lift Coefficient (C_L) vs. Time

FigureS 1–4 show the lift coefficient as a function of time for Reynolds numbers: $Re = [100, 200, 300, 400]$. The transient and periodic regions are identified using a threshold based on the maximum lift coefficient in the 5-second time period. The threshold was set to be the time interval prior to the lift coefficient reaching 97.5% of its maximum value. This value was chosen because when in its periodic region, the C_L value would fluctuate to below 2.5%.

B. Discussion

Table 1 Average oscillation characteristics between Reynolds numbers.

Re	Transient Cutoff (s)	Period (s)	Frequency (Hz)	Amplitude	$C_{L,\max}$
100	3.44	0.3213	3.1120	1.9216	1.9349
200	3.26	0.2988	3.3469	2.4149	2.4434
300	1.82	0.2913	3.4331	2.0245	2.0453
400	1.58	0.2845	3.5149	2.1441	2.1660

Table 1 shows the average oscillation characteristics after the transient threshold for the full 20 seconds. The average period, frequency, and amplitudes were calculated using the peaks of each cycle in the periodic region.

As the Reynolds number increases, the transient region decreases, suggesting that the flow reaches its oscillatory behaiver quicker at higher Re. In this dataset, the frequency consistently increases with Reynolds number, while amplitude and $C_{L,\max}$ are all over the place. The periodic amplitude remains generally the same C_L of 2.

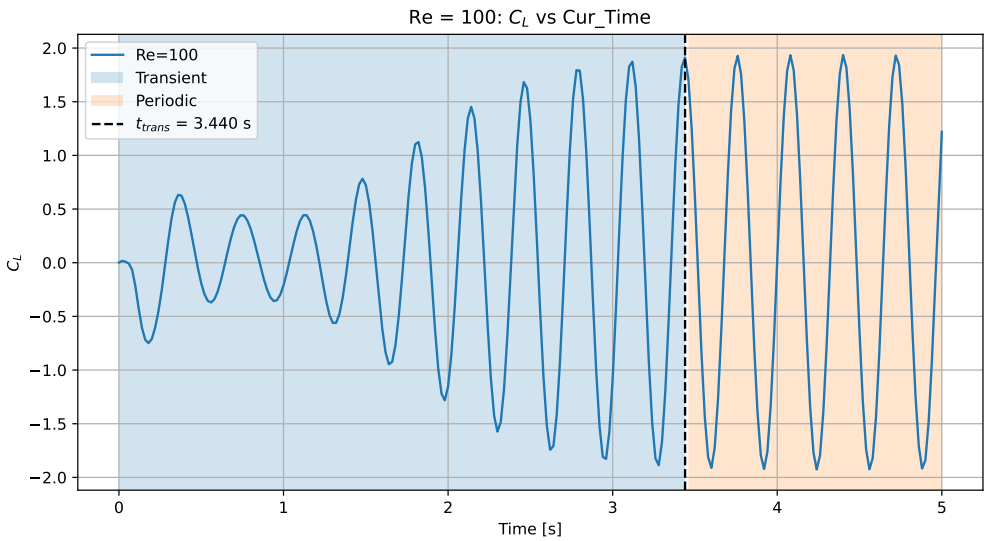

Fig. 1 C_L **vs time for** $Re = 100$**.**

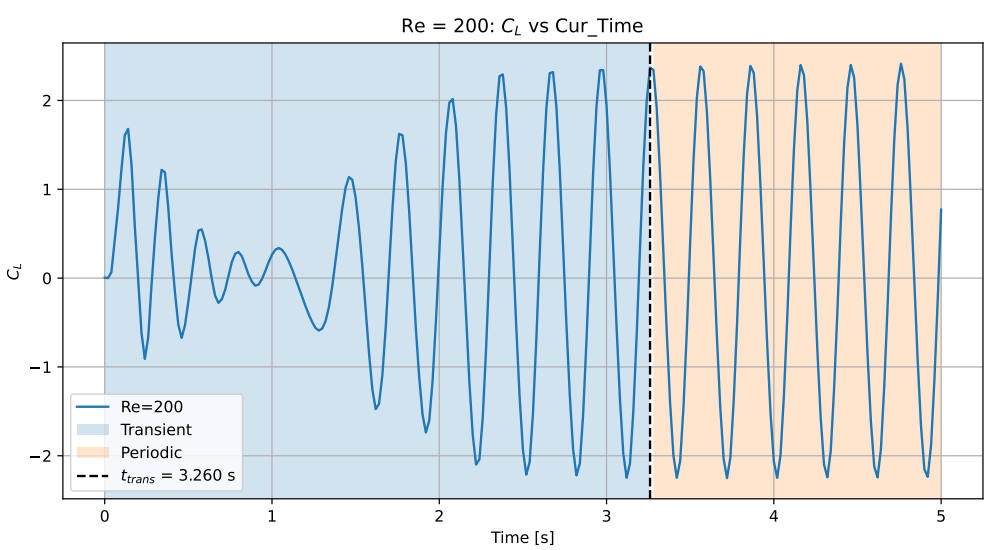

Fig. 2 C_L **vs time for** $Re = 200$**.**

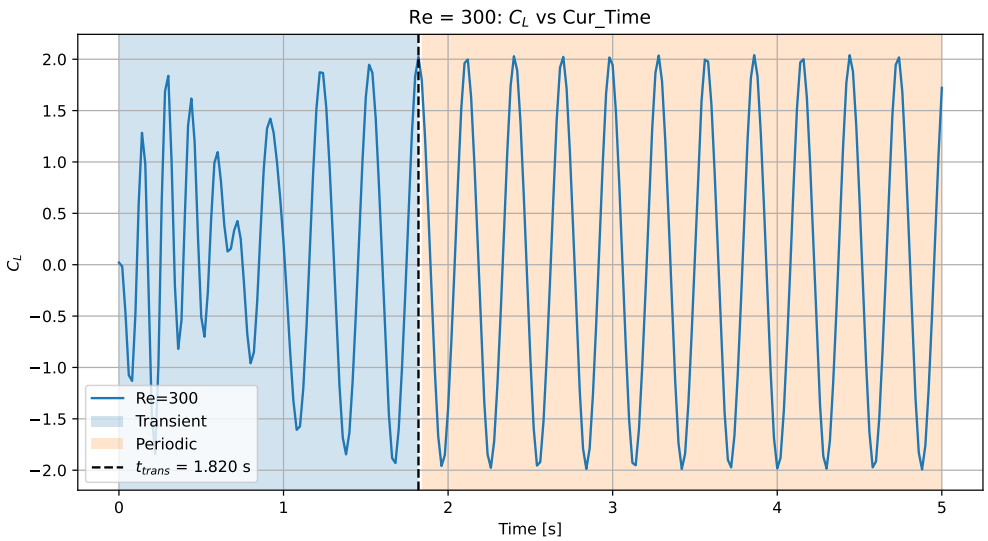

Fig. 3 C_L **vs time for** $Re = 300$**.**

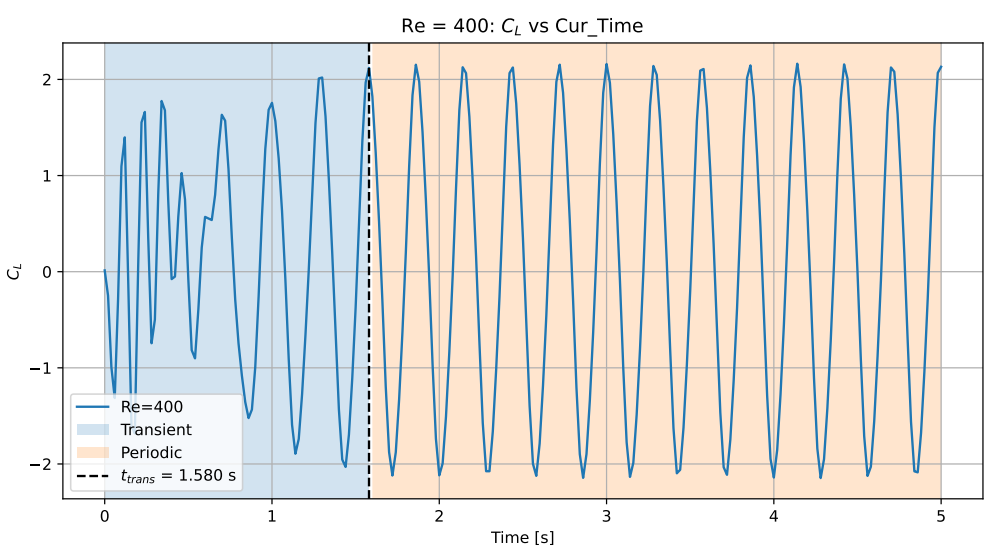

Fig. 4 C_L **vs time for** $Re = 400$**.**

II. PSD and Strouhal Number

A. Power Spectral Density

Figures 5–8 show the power spectral density of the lift coefficient fluctuations computed using Welch's method with constants NFFT = 512 and f_s = 50 Hz.

B. Strouhal Number

The dominant frequency was identified as the frequency corresponding to the peak of the PSD. The Strouhal number was calculated using the equation:

$$St = \frac{fD}{U},$$

where f is the dominant frequency obtained from the PSD, D is the characteristic length (2 m), and U is the freestream velocity (34.03 m/s).

Table 2 Dominant oscillation frequency from PSD analysis and corresponding Strouhal number as a function of Reynolds number.

Re	f_{PSD} (Hz)	St
100	3.1250	0.1837
200	3.3203	0.1951
300	3.4180	0.2009
400	3.5156	0.2066

C. Comparison to prior research

The overall trend of the Strouhal number with Reynolds number is consistent with prior research, showing a gradual increase in St as Re increases. At $Re = 100$, the present result ($St = 0.1837$) is slightly higher than the reported value of approximately 0.166, with differences potentially due to numerical setup and resolution.

At $Re = 200$, the flow is in a transitional regime, and the literature reports a wide spread of values ($St \approx 0.175$ to 0.195). The computed value of $St = 0.1951$ lies near the upper end of this range and is close to the 2-D Posdziech result.

For $Re = 300$ and $Re = 400$, the agreement improves. The computed values ($St = 0.2009$ and 0.2066, respectively) closely match reported trends, with prior research values near 0.205 and 0.235. Overall, the results demonstrate good qualitative agreement and reasonable quantitative accuracy, with discrepancies likely attributable to modeling assumptions such as two-dimensional flow and numerical resolution.

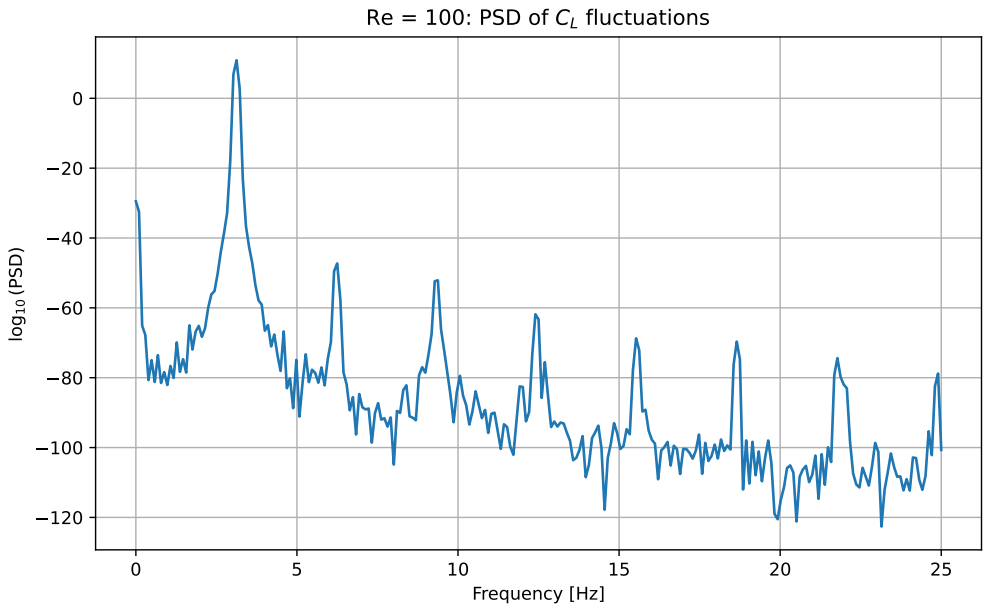

Fig. 5 PSD of C_L fluctuations for $Re = 100$.

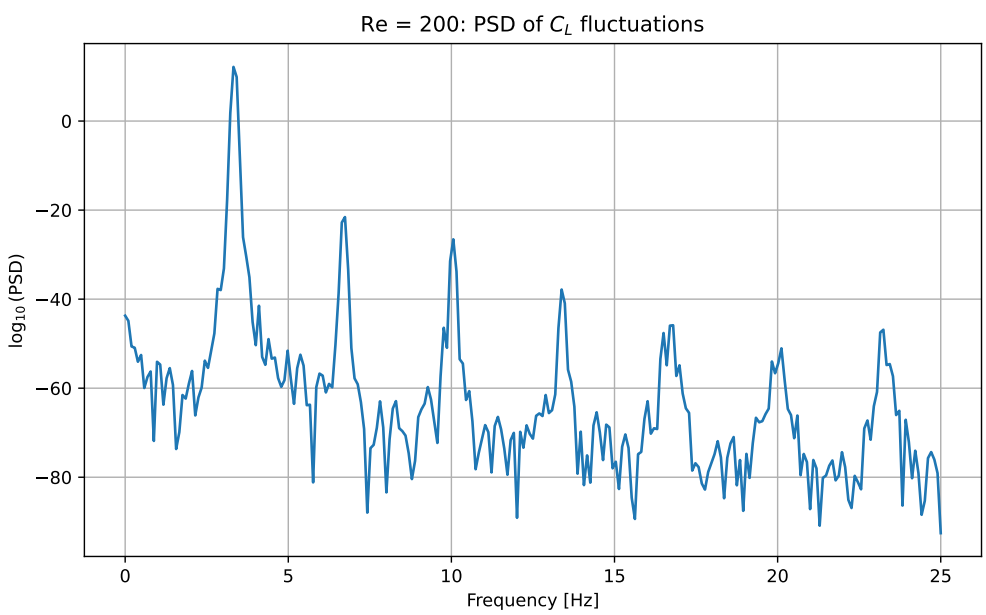

Fig. 6 PSD of C_L fluctuations for $Re = 200$.

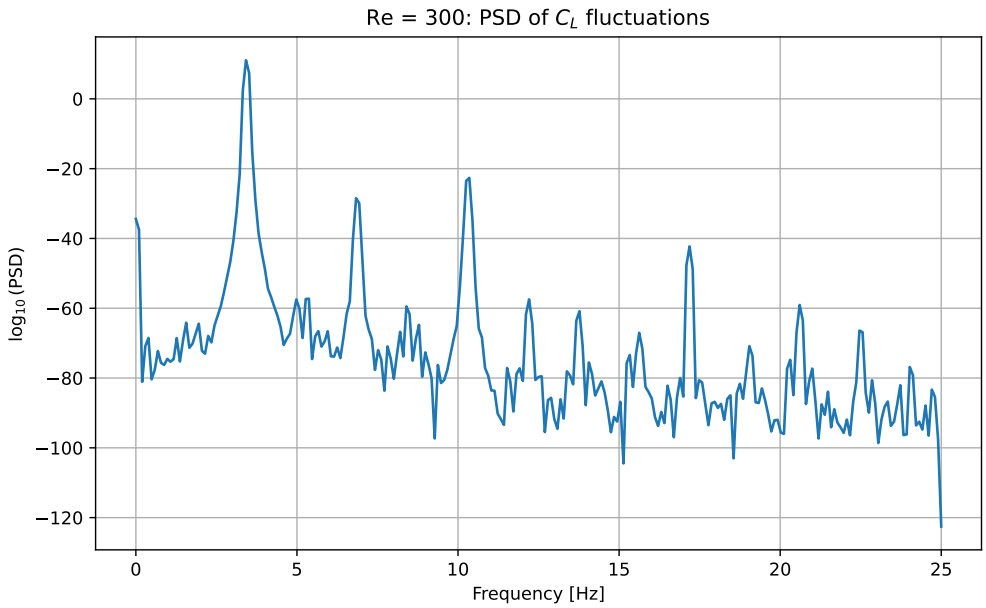

Fig. 7 PSD of C_L fluctuations for $Re = 300$.

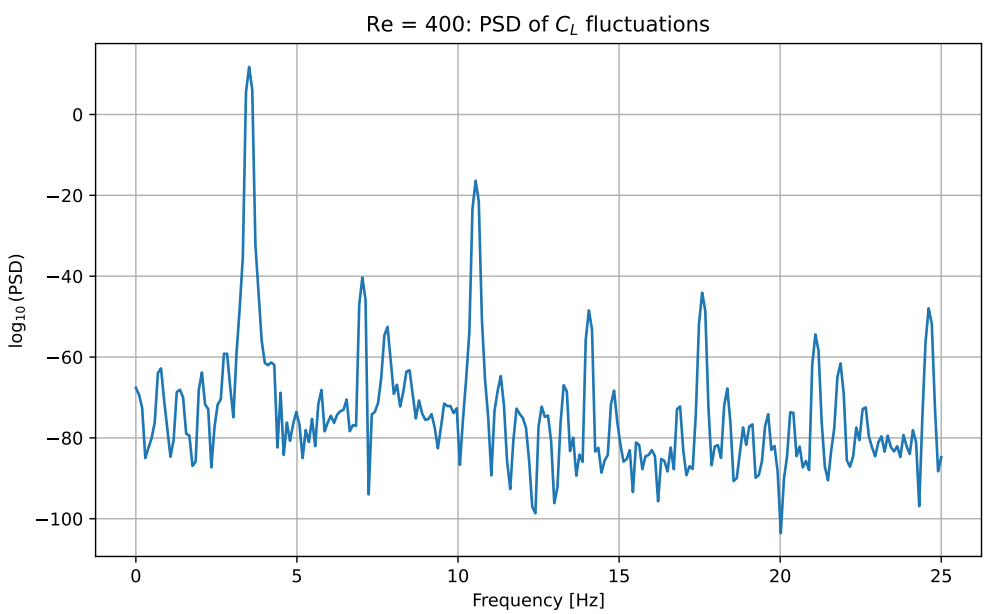

Fig. 8 PSD of C_L fluctuations for $Re = 400$.

III. Flow Field Visualization

A. Snapshots from Paraview

Figures 9–12 show the velocity field for each Reynolds number at their corresponding $C_{L,\max}$ time.

B. Estimates of vertex spacing

The spacing was measured using the ruler tool in Paraview. The distance between the leading edges of two vortexes shed from the same side of the cylinder were measured, as well as the approximate distance between the centers of the vortexes. These values were then averaged to produce the results in Table 3.

Table 3 Average vortex spacing measured from flow visualization at different Reynolds numbers.

Re	Spacing 1 (m)	Spacing 2 (m)	Average Spacing (m)
100	8.6435	8.6132	8.6284
200	7.7391	7.7994	7.7693
300	8.0111	7.9527	7.9819
400	7.3160	7.4974	7.4067

IV. Conclusion

The results demonstrate that as Reynolds number increases, the flow transitions more quickly to periodic behavior and exhibits higher-frequency vortex shedding. This solution matches with previous research on the topic. The flow field visualizations confirm the presence of unsteady wake structures associated with vortex shedding.

Appendix

The GitHub repository containing all simulation files and post-processing scripts is available at:
`https://github.com/nikitaaa27/AE462-git/lab5`

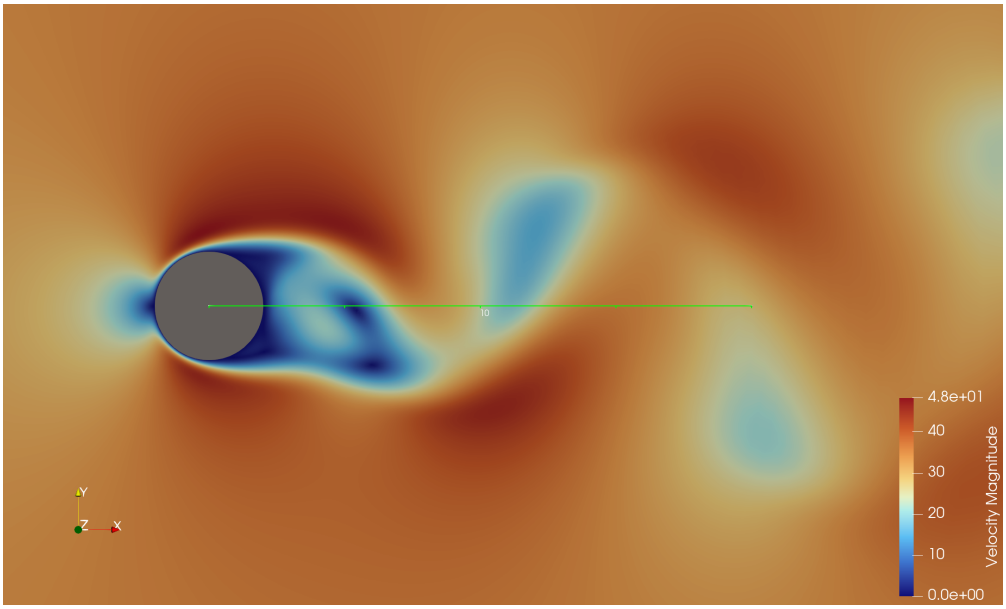

Fig. 9 Velocity field for $Re = 100$.

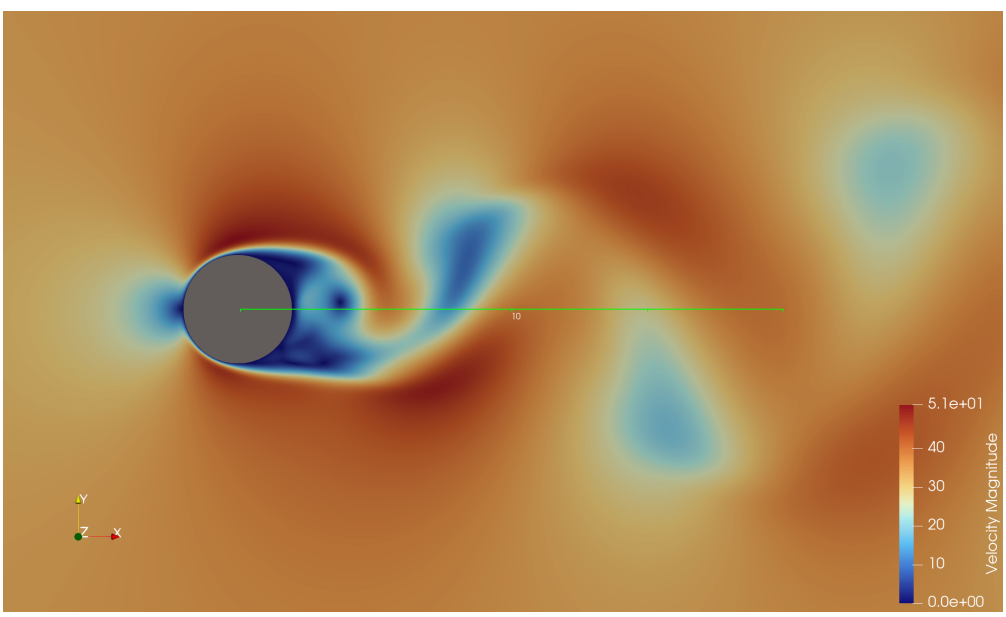

Fig. 10 Velocity field for $Re = 200$.

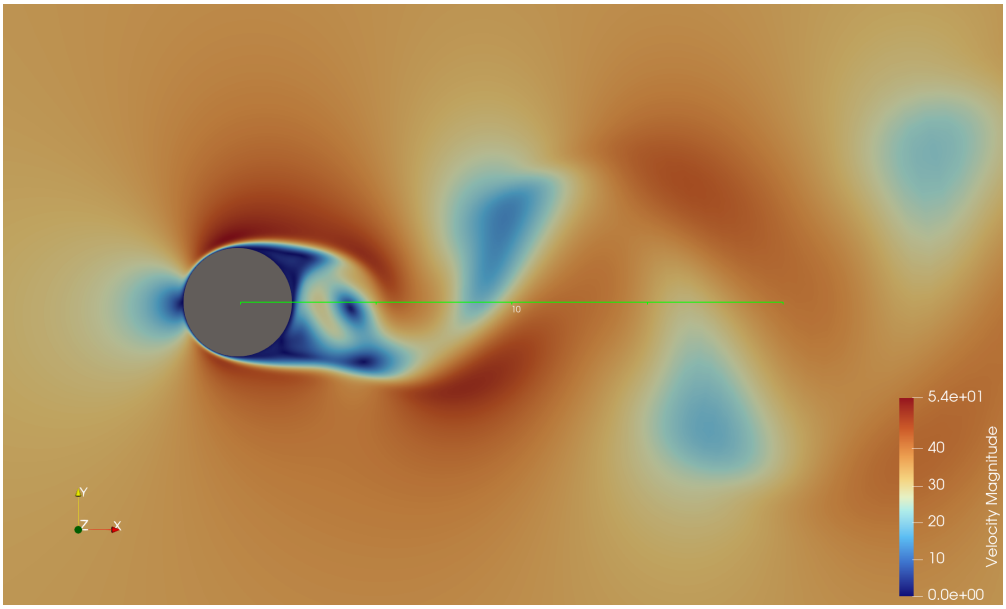

Fig. 11 Velocity field for $Re = 300$.

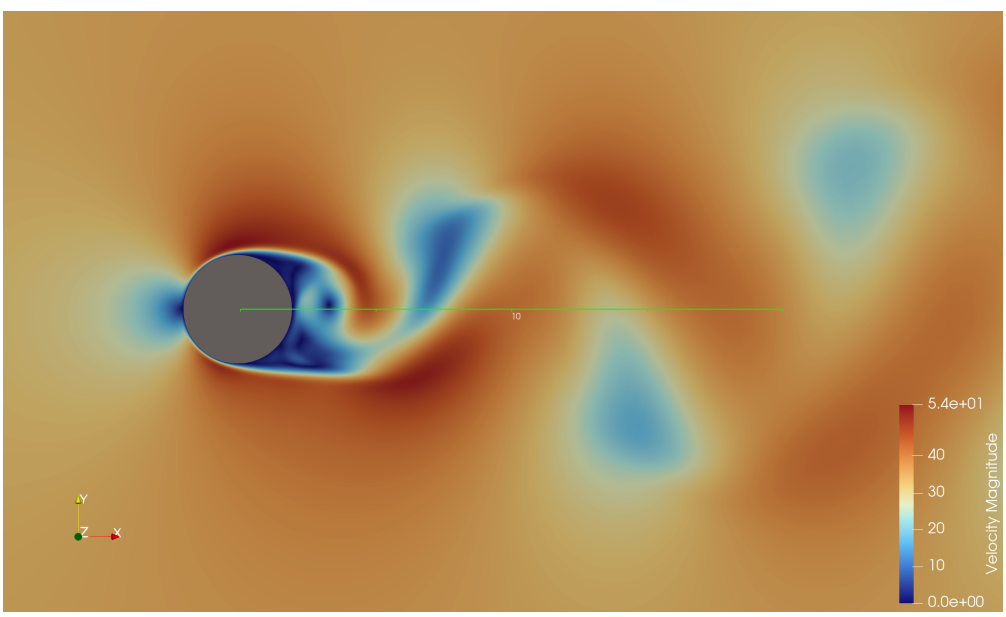

Fig. 12 Velocity field for $Re = 400$.